

Seeing the Same Task Differently[8pt] Vision-Language Model Inclusion Criteria

Technical report from the repository evaluation outputs

Chapter 1

Model Inclusion Criteria

1.1 Purpose of the Selection

This study evaluates a deliberately bounded set of vision-language models. It does not attempt to reproduce a full public leaderboard or catalogue every model available at the time of execution. The purpose is narrower: to compare a small group of practically accessible, general-purpose multimodal systems under one benchmark implementation.

The model choice was informed by two contemporary vision-language leaderboards: MMMU-Pro¹ and Arena Vision Overall². The MMMU-Pro table below uses the public LLM Stats MMMU-Pro leaderboard snapshot.³ These leaderboards identify relevant model families; the repository artifacts determine which models can actually be compared in this study.

The final cohort contains six models:

- **OpenAI:** GPT-4o, GPT-4.1, GPT-5;
- **Anthropic:** Claude Sonnet 4.5;
- **Google:** Gemini 3.5 Flash;
- **Alibaba Cloud:** Qwen 3.5.

This selection gives extra depth to the GPT family because it is the strongest and most consistently relevant family for this study: GPT models appear repeatedly near the top of the leaderboard evidence, and the repository contains multiple completed GPT-family artifacts under the same benchmark pipeline. The other major families are represented by one model each, which keeps the comparison broad without turning the study into a full leaderboard survey.

1.2 Why These Two Leaderboards Were Used

MMMU-Pro and Arena Vision Overall were chosen because they provide complementary views of current vision-language performance. MMMU-Pro emphasizes structured multimodal reasoning, while Arena Vision Overall reflects broader comparative model behavior across vision-language tasks. Using both reduces dependence on a single benchmark source while keeping the selection process compact.

¹<https://artificialanalysis.ai/evaluations/mmmu-pro>, accessed 2026-06-11.

²<https://arena.ai/leaderboard/vision>, accessed 2026-06-11.

³<https://llm-stats.com/benchmarks/mmmu-pro>, accessed 2026-06-11.

The leaderboard excerpts in Tables 1.1 and 1.2 are fixed snapshots recorded at the time of model selection. Each table contains 25 entries: enough to show the main families visible near the top of each ranking, but not so many that the inclusion argument becomes a leaderboard survey.

1.3 MMMU-Pro Leaderboard Context

Table 1.1 presents the MMMU-Pro leaderboard context used during model selection. It shows that GPT, Gemini, Qwen, Claude, and Muse Spark all belong to the relevant upper part of the model landscape.

Model name	Rank	Score	Family	Company
Gemini 3.5 Flash	1	0.836	Gemini	Google
GPT-5.5	2	0.832	GPT	OpenAI
Gemini 3 Flash	3	0.812	Gemini	Google
GPT-5.4	3	0.812	GPT	OpenAI
Gemini 3 Pro	5	0.810	Gemini	Google
Gemini 3.1 Pro	6	0.805	Gemini	Google
Muse Spark	7	0.804	Muse Spark	Meta
Kimi K2.6	8	0.801	Kimi	Moonshot AI
GPT-5.2	9	0.795	GPT	OpenAI
Qwen3.6 Plus	10	0.788	Qwen	Alibaba Cloud / Qwen Team
Kimi K2.5	11	0.785	Kimi	Moonshot AI
GPT-5	12	0.784	GPT	OpenAI
MiniMax M3	13	0.781	MiniMax	MiniMax
MiMo-V2.5	14	0.779	MiMo	Xiaomi
Claude Opus 4.6	15	0.773	Claude	Anthropic
Gemma 4 31B	16	0.769	Gemma	Google
Qwen3.5-122B-A10B	16	0.769	Qwen	Alibaba Cloud / Qwen Team
Gemini 3.1 Flash-Lite	18	0.768	Gemini	Google
GPT-5.4 mini	19	0.766	GPT	OpenAI
o3	20	0.764	o-series	OpenAI
GPT-5.5 Instant	21	0.760	GPT	OpenAI
Qwen3.6-27B	22	0.758	Qwen	Alibaba Cloud / Qwen Team
Claude Sonnet 4.6	23	0.756	Claude	Anthropic
Qwen3.6-35B-A3B	24	0.753	Qwen	Alibaba Cloud / Qwen Team
Qwen3.5-35B-A3B	25	0.751	Qwen	Alibaba Cloud / Qwen Team

Table 1.1: Twenty-five-entry MMMU-Pro leaderboard excerpt used during model selection. The table defines part of the broader candidate landscape; it does not determine automatic inclusion in the main study.

1.4 Arena Vision Leaderboard Context

Table 1.2 presents the Arena Vision Overall context. It reinforces the relevance of Claude, Gemini, GPT, Qwen, and Muse Spark as candidate families for a vision-language study.

Model name	Rank	Score	Family	Company
claude-opus-4-7-thinking	1	1309 $\pm$ 8	Claude Opus	Anthropic
claude-fable-5	2	1307 $\pm$ 15	Claude	Anthropic
claude-opus-4-6-thinking	3	1301 $\pm$ 8	Claude Opus	Anthropic
claude-opus-4-7	4	1301 $\pm$ 8	Claude Opus	Anthropic
claude-opus-4-6	5	1297 $\pm$ 7	Claude Opus	Anthropic
muse-spark	6	1295 $\pm$ 10	Muse Spark	Meta
gemini-3-pro	7	1290 $\pm$ 8	Gemini	Google
claude-opus-4-8-thinking	8	1289 $\pm$ 11	Claude Opus	Anthropic
gpt-5.5-high	9	1284 $\pm$ 8	GPT	OpenAI
gpt-5.4-high	10	1283 $\pm$ 8	GPT	OpenAI
gpt-5.5	11	1283 $\pm$ 8	GPT	OpenAI
claude-opus-4-8	12	1280 $\pm$ 11	Claude Opus	Anthropic
gemini-3.1-pro-preview	13	1280 $\pm$ 6	Gemini	Google
gpt-5.2-chat-latest-20260210	14	1279 $\pm$ 7	GPT	OpenAI
claude-sonnet-4-6	15	1278 $\pm$ 7	Claude Sonnet	Anthropic
gpt-5.5-instant	16	1276 $\pm$ 9	GPT	OpenAI
gpt-5.4	17	1275 $\pm$ 8	GPT	OpenAI
gemini-3-flash	18	1272 $\pm$ 6	Gemini	Google
kimi-k2.6	19	1264 $\pm$ 8	Kimi	Moonshot
qwen3.7-plus-preview	20	1263 $\pm$ 11	Qwen	Alibaba
dola-seed-2.0-pro	21	1259 $\pm$ 8	Seed	ByteDance
gemini-3-flash (thinking-minimal)	22	1256 $\pm$ 6	Gemini	Google
gpt-5.4-mini-high	23	1256 $\pm$ 7	GPT	OpenAI
gemma-4-31b	24	1252 $\pm$ 7	Gemma	Google
grok-4.20-beta-0309-reasoning	25	1251 $\pm$ 7	Grok	xAI

Table 1.2: Twenty-five-entry Arena Vision Overall leaderboard excerpt used during model selection. The table documents the broader candidate landscape; final inclusion still requires completed repository artifacts and compatibility with the common evaluation procedure.

1.5 From Leaderboard Families to Final Cohort

The final cohort is not a top-(k) slice of either leaderboard. It is the intersection of three constraints: relevance in the broader model landscape, compatibility with the common benchmark pipeline, and the existence of completed repository artifacts.

The family-level screening proceeds as follows. The leaderboards identify five main candidate families for this study: GPT, Gemini, Claude, Qwen, and Muse Spark. Muse Spark is excluded despite its high rank because it is not generally available through a reproducible public API route. Meta describes Muse Spark as available at `meta.ai` and the Meta AI app, with API access offered

only as a private preview to selected users.⁴ Artificial Analysis likewise reports that there are no currently benchmarked API providers for Muse Spark.⁵

After excluding Muse Spark, the remaining families are GPT, Gemini, Claude, and Qwen. GPT receives three slots because it is the strongest family for a within-family analysis in this study: it appears repeatedly near the top of both leaderboards and the repository contains completed artifacts for several GPT generations. The other three families receive one model each. This keeps the comparison broad across providers while preserving a controlled GPT-family trajectory.

This gives the following six-model cohort:

Model	Provider	Reason for inclusion
GPT-4o	OpenAI	Serves as the established GPT-family multimodal reference point and anchors the within-family comparison.
GPT-4.1	OpenAI	Provides an intermediate GPT generation, allowing changes across OpenAI models to be studied across tasks rather than inferred from a single pair.
GPT-5	OpenAI	Provides the latest GPT model evaluated in the cohort and completes the three-model GPT-family comparison.
Claude Sonnet 4.5	Anthropic	Represents the Claude family with a strong general-purpose multimodal model for cross-provider comparison.
Gemini 3.5 Flash	Google	Represents the Gemini family with a high-performing and practically usable model that is especially relevant given Gemini’s strong leaderboard presence.
Qwen 3.5	Alibaba Cloud	Represents the Qwen family and adds a non-US provider family to the main comparison.

Table 1.3: Models included in the main comparison.

The cohort is intentionally asymmetric. Equal provider counts would remove the GPT generational comparison, which is one of the useful internal controls of the study. Instead, the cohort uses a (3+1+1+1) design: three GPT models for depth, and one Claude, one Gemini, and one Qwen model for breadth.

1.6 Why Additional Leaderboard Models Were Not Included

Many models visible in the leaderboard context were not included in the main cohort. This does not mean they are unimportant. It means that including them would have changed the study from a controlled repository-backed comparison into a broader leaderboard survey.

In particular, additional GPT, Claude, Gemini, Qwen, Kimi, Gemma, Grok, MiMo, Seed, MiniMax, and related systems could have been considered. They were not automatically included because the study requires completed artifacts from the same benchmark pipeline and a comparison setting close enough for the reported results to be interpreted together.

The final cohort should therefore be read as a deliberately measured subset of a larger model landscape. The leaderboards establish family relevance; the repository artifacts determine model inclusion.

⁴<https://ai.meta.com/blog/introducing-muse-spark-msl/>, accessed 2026-06-11.

⁵<https://artificialanalysis.ai/models/muse-spark/providers>, accessed 2026-06-11.

1.7 Fine-Tuning Case Study

Qwen2.5-VL-3B is included separately in a Fashion-MNIST fine-tuning case study. It is not treated as another member of the main model ranking. The purpose of this experiment is different: it measures whether parameter-efficient fine-tuning can improve a compact model on a defined visual classification task.

The case study compares the same Qwen2.5-VL-3B base model before and after LoRA fine-tuning. This paired design controls for model architecture and scale, so the observed difference can be attributed more directly to task-specific adaptation.

Keeping this case study separate is necessary for a fair interpretation. Its model size, deployment method, training exposure, and task specialization differ from those of the general-purpose models in the main cohort. Its results therefore support conclusions about fine-tuning effectiveness on Fashion-MNIST, not a general claim that Qwen2.5-VL-3B is directly superior or inferior to the six-model comparison cohort.

1.8 Scope of the Conclusions

The selected models support three forms of analysis:

- comparison of general-purpose multimodal systems across OpenAI, Anthropic, Google, and Alibaba Cloud;
- comparison of successive GPT-family models under the same benchmark protocol;
- comparison of a compact base model with its task-specific fine-tuned counterpart in a separate case study.

Accordingly, conclusions are limited to the evaluated model versions, endpoints, prompts, benchmark samples, and execution settings. The leaderboard tables document the broader model landscape and justify why the selected families were reasonable candidates. They do not override the stricter inclusion rule that reported comparisons must be traceable to completed artifacts produced by the common evaluation pipeline.